Harsh Makhija
Intern

IIMBx Tools & Impact Report

Tools Evaluated and Quarterly Impact Summary (April – June 2026)

Tools Checked & Evaluated

Antigravity (Google)
Coding agent used for building dashboards, websites, and prototypes. Worked well when given visual references/drawings but hit frequent rate-limit issues, pushing exploration of alternatives.

Claude (Opus / Claude Max / Claude Fable 5)
Used for video editing (auto captions/music), image-based reasoning (interpreting hand-drawn wireframes), and later evaluated for safer agent workflows using git-branching so changes can be reviewed before going live.

Gemini
Used as the core AI model for the PM dashboard assistant and course suggestion engine. Also flagged as the strongest model for creative output like social media posts and flyers.

NotebookLM (Google)
Evaluated as a free, zero-setup alternative to DeepTutor for studying and course support. Recommended to the team over DeepTutor for faculty use.

DeepTutor
Locally-hosted study tool that wraps around existing AI models (Claude/GPT/Gemini) rather than having its own AI. Tested and ultimately not recommended — too costly and requires heavy technical setup (Python, Node.js, manual restarts).

Ai.cofounder
AI tool for market research and idea validation; pulls data from Reddit, Instagram, YouTube, and competitor sites to generate market-ready reports. Tested via free trial for validating market gaps.

Manus.ai
Alternative AI page-design tool tested as a comparison against the in-house website agent for replicating and designing pages.

Comet
Briefly evaluated as a possible alternative to Antigravity during a period of rate-limit issues.

ExaAi (MCP)
Tested as an add-on integration inside the dashboard to see if it added value.

Stitch (MCP)
Tested for website generation; found to work well for websites but made dashboards overly aesthetic rather than functional.

CrewAI
Flagged for future exploration as a multi-agent framework.

Codex (via ChatGPT)
Evaluated for automating the YouTube upload workflow — handling titles, descriptions, thumbnails, tags, and scheduling for both Shorts and long-form videos.

OmniDimension
Voice agent platform used for outbound/inbound calling; iterated on prompts and Excel data templates to fix data population and language-handling issues.

OmniRoute
Tested as an agent/routing tool; findings shared via demo videos.

Groq
Used as a free API provider to power the course suggestion engine.

Freedcamp API
Backbone of the PM dashboard; pulled task, milestone, and team data for the whole tracking system.

Google Lighthouse CLI
Integrated into the website agent to automatically score every generated page on performance, SEO, and UX.

Railway Cloud
Briefly evaluated as a hosting option during early PM dashboard deployment.

Render
Used as the primary hosting platform for the PM dashboard, CEO dashboard, and suggestion engine.

Microsoft MarkItDown
Referenced/shared as a potential document-conversion tool.

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Quarterly Impact Report

April – June 2026

April: Core Systems Foundation

PM Dashboard — Built from Scratch on the Freedcamp API

The team was struggling to track project status and milestones across people and teams with no central view. Built a complete PM dashboard directly on top of the Freedcamp API to solve this — instant visibility into overdue tasks, milestones, and team-wise workload through a single “Hawkeye” view. Embedded a Gemini-powered AI assistant directly into the dashboard so team members could query project status without digging through Freedcamp manually. This system is still actively used by the team today and was purpose-built for organizational agility.

Website Redesign — First Prototypes

Designed and shipped the first set of visual prototypes for the new IIMBx website, reverse-engineering strong reference sites to get the wireframes right, laying the groundwork for the full redesign that followed.

May: Executive Reporting & Flagship AI System

CEO Dashboard — Real-Time Reporting Across Every Program

Built and launched a CEO-level dashboard pulling data directly from Google Sheets across every program IIMBx runs — BBA DBE, Swayam, and others. Every KPI across production and course performance is tracked in real time in one place, replacing the slow, manual Power BI reporting process that existed before. This gave leadership a single, always-current view instead of waiting on manually assembled reports.

Website Agent — Autonomous Design & Build System

Rebuilt an existing legacy agent into a genuinely production-grade system: feed it any webpage URL, and it reads the page, generates four distinct prototype designs against brand guidelines, and hosts them live on the front end for review — complete with visible, ready-to-use code for each version. This turned website redesign from a manual design-and-build cycle into an automated, on-demand pipeline. Both the frontend and the underlying agent architecture were built out from the ground up — a significant engineering effort on its own.

June: Scaling, Quality Control & Tool Evaluation

Automated Quality Grading

Added Google Lighthouse CLI scoring directly into the website agent, so every AI-generated page is automatically checked for performance, SEO, and user experience before going live — no manual audit needed.

Agent Architecture Simplified for Reliability

Re-engineered the website agent’s internal structure from over 20 separate agents down to 3 core agents plus one manager agent, making the system leaner, faster, and easier to maintain.

Tool Evaluation & Cost Optimization

Ran hands-on evaluations of emerging AI tools (including DeepTutor, NotebookLM, Manus) to find the right fit for faculty and course support, with findings and recommendations reported back to the team. Also pioneered a per-folder skills.md structuring approach that measurably cut AI processing costs and reduced code bloat across the system.